M T W T F S S
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MC :- UT2 Question Bank

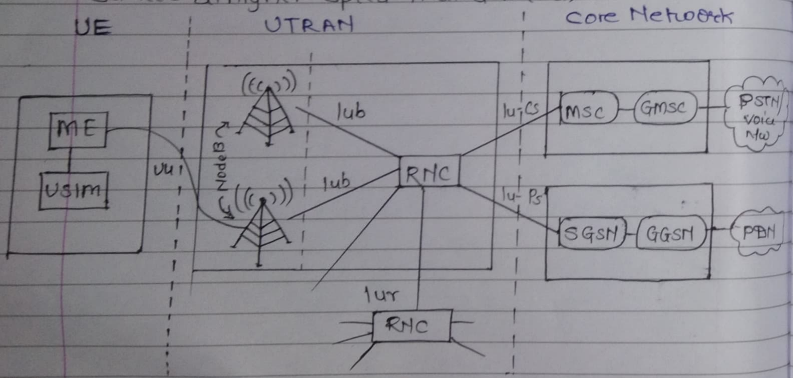
✓ 1) Describe Bluetooth Architecture.
✓ 2) Explain UMTS architecture
✓ 3) Explain WLAN & state advantages & disadvantages of WLAN.
4) Write short note on: Macro Mobility.
5) Discuss in detail about Wi-fi security protocol.
✓ 6) Write short note on: LTE.
✓ 7) Explain system architecture of IEEE 802.11
8) Explain HIPERLAN 2.

14/4/26

* Answers for UT2 QB :-

Q] Explain UMTS Architecture

- UMTS [Universal Mobile Telecommunication system] is a 3G Mobile Communication system standardized by 3rd Generation Partnership Project (3GPP).
- UMTS doesnot define a complete new 3G system rather it specifies a smooth transition from second generation GSM or TDMA systems to the third generation.
- Supports Circuit switched and packet switched transmissions.
- It was designed to provide voice, data and multimedia services at higher speed that GSM(2G).



- UMTS Architecture is divided into 3 main parts
  1. User Equipment (UE)
  2. UTRAN [UMTS Terrestrial Radio Access Network]
  3. Core Network (CN)

1. User Equipment (UE) :-
This is the mobile device used by sim
• components : ME & USIM

- ME [Mobile Equipment] : The Radio terminal connecting to the radio Interface using Uu interface.
- USIM [Universal Subscriber Identity Module] : Store subscriber Identity, Authentication algorithms, Encryption key, etc.
- function :
  • Communication with the radio network.
  • Encryption and authentication
  • Running user application.

2. UTRAN [UMTS Terrestrial Radio Access Network) :-
UTRAN connect the Mobile devices to the core Network.
• components : Node B, RNCs, Interface.

- Node B :
  • Equivalent to GSM's BTS (Base Transceiver station)
  • Handles radio transmission and reception.
  • Manages power control and spreading code (WCDMA).
- RNCs [Radio Network controller] :
  • Control one or More Node Bs.
  • Manages :
    - Radio resource control
    - Handover management
    - Load Control
    - System information broadcasting
    - Radio channel ciphering & driciphering
    - Congestion control.
    - Encryption
  • RNC is connected to MSC and SGSN to route Circuit Switched (CS) and Packet Switched (PS) data.

- Interfaces !
  - Uu - Betn UE and Node B.
    comparable to Um interface in GSM.
  - Iub - Between Node B and RNC
  - Iur - Between RNCs
  - Iu - Betn RNCs and Core Network
    similar to A interface in GSM.

3. Core Network (CN) :—
Core Network connects users to other network
- component : MSC, GMSC, SGSN and GGSN.

- Circuit switched (CS) domain
  - Handles voice call
  Main Element
  - MSC (Mobile Switching Center) - Call switching
  - VLR (Visitor Location Register) - Temporary subscriber ~~of GMSC (Gateway MSC)~~
  - GMSC (Gateway MSC) - Connects to external network (GMSC)

- Packet-Switched (PS) Domain
  - Handles data science (internet, email, etc
  - ~~GSSN~~ Main Element
    SGSN (Serving GPRS Support Node)

previations :—

MSC - Mobile Switching Center	GMSC - Gateway MSC
SGSN - Serving GPRS Support node	GGSN - Gateway GSN

PSTN - Public Switch telephone network
PDN - Packet Data Network.

- When users wants to call then RNCs connected MSC and GMSC through Iu-Cs (Circuit Switch) Interface then GMSC connect with PSTN.
- When user wants to use data then RNCs connected with SGSN and GGSN via Iu-Ps (Packet Switch) Interface after that GGSN ~~packet switc~~ is connect with PDN.

Q7 Explain WLAN and State advantages & disadvantages of W-LAN.

→ Local Area Network : It is Private Network that connects computers, devices within a limited area like building, office, campus, etc.

- Wireless Local Area Network : It is a Wireless Computer network that links 2 or more devices using Wireless (radio) communication to form a LAN within a limited area such as school, building, office, etc.
- The IEEE 802.11 group of standards specify the technologies for wireless LANs.
- Home users can create a wireless network out of an existing wired network and wirelessly extend the reach of the Internet throughout the home on Multiple computers.
- By using WLAN, it is now not required that every workstation and conference room be wired up to hubs and switches with cables.

- Advantages :—
1. Flexibility : Within radio coverage, a user can easily communicate without any restriction.
2. Simplified planning : Wireless adhoc networks do not require any planning for configuring a network.

3. (Almost) no wiring difficulties : Because no wiring is required, it can be easily installed where wiring is difficult (eg., historic buildings, firewalls)
4. Robust : Wireless LAN is more robust against disasters like eg., Earth quake, fire or users pulling a plug.
5. Cost effective : Once a wireless network is installed, adding new additional user will not increase further cost.

• Disadvantages :—

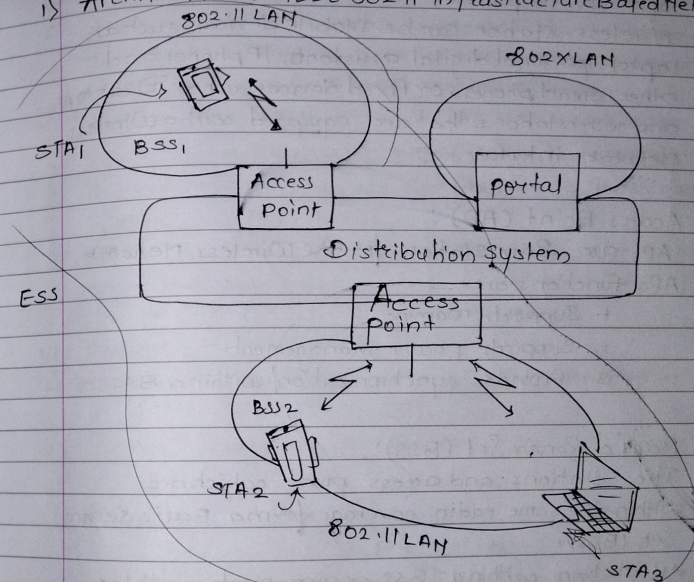
1. Lower Bandwidth and transmission quality : Wireless LAN offers very low bandwidth (1-10 Mbit/s) compared to wired networks due to shared medium. Also it has high error rates due to interference. Hence it offers low QOS as compared to wired network.
2. Many Proprietary Solutions : (data transfer rate)
   ↓↑
   ↑es devices connect ∝ rate ↓es
3. Local regulatory restrictions : Several countries impose different spectral restrictions. Due to this it is difficult to establish global WLAN solutions.
4. Lower safety & security : Security concerns are high in wireless networks. The open radio interface makes eavesdropping much easier in WLANs than wired network.

Q] Explain system Architecture of IEEE 802.11.

→ IEEE 802.11 System Architecture :-

- IEEE ~~208~~ 802.11 LAN can be configured as "Infrastructure Based network" or as "ad-hoc Network."

1) Architecture of IEEE 802.11 Infrastructure Based Network



Note:- Take SS from video for this ans

I] Station (STA):
- Several nodes, called Stations (STA), are connected to an Access points.
- All stations are equipped with Wireless Network Interface cards (WNICs) and contain functionality of the 802.11 protocol.
- Wireless station can be Mobile devices such as Laptop, personal digital assistants, IP phones and other smart phones, or fixed devices such as Desktops and workstations that are equipped with a Wireless Network Interface.
-

II] Access Point (AP):
- APs are Base station for the Wireless Network.
- APs functions are:
 1. Supports roaming
 2. Supports power management
 3. Provide synchronization within a BSS

III] Basic Service Set (BSS):
- The stations and access point which are within the same radio coverage form a Basic Service Set (BSS).
- All station within a BSS communicate with the same ~~po~~ access point & compete for shared medium
- Access point can be connected to another access points via Distribution System.

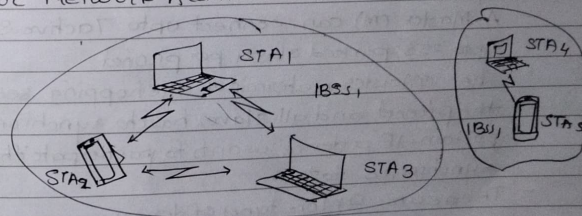
IV] Extended Service System (ESS):
- A set of connected BSSs together form ESS via AP to form a single network and thereby extends the Wireless Coverage Area.

- This network is now called Extended Service Set (ESS).
- ~~A distribution system co~~ Every ESS has its own identifier called the ESSID which is 32 byte (max) character string

V] Distribution Systems (DS):
- A distribution system connects BSSs via the AP.
- The distribution system connects the wireless networks via APs with a Portal, which forms the interworking unit to other LANs
- works as a Backbone network & handles data transfer.
- The architecture of the distribution system is not specified further in IEEE 802.11
- It could consist of bridged IEEE LANs, wireless links, or any other networks.

VI] Portal
- acts as a Internet working unit to connect other LANs.

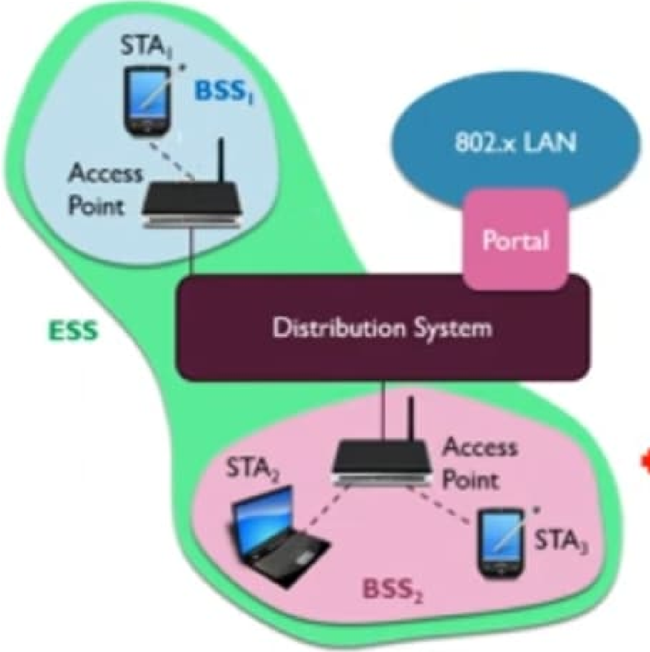
2) Ad-hoc Network Architecture.



- In addition to infrastructure-based networks, IEEE 802.11 allows the building of ad-hoc networks between stations.
- Every ad-hoc network thus forming one or more Independent BSSs (IBSS) as shown in fig.

SYSTEM ARCHITECTURE

- Figure shows the architecture
  - Several nodes, called stations (STA_i), are connected to access points (AP).
  - Stations are terminals with access mechanisms to the wireless medium and radio contact to the AP.
  - The stations and the AP which are within the same radio coverage form a basic service set (BSS_i).
  - The example shows two BSSs – BSS1 and BSS2 – which are connected via a distribution system.
  - A distribution system connects several BSSs via the AP to form a single network and thereby extends the wireless coverage area.
  - This network is now called an extended service set (ESS).
  - The distribution system connects the wireless networks via the APs with a portal, which forms the interworking unit to other LANs.



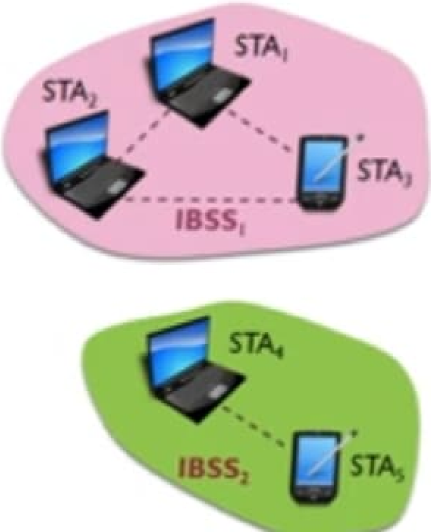
SYSTEM ARCHITECTURE

- Every ESS has its own identifier, the ESSID.
- The ESSID is the 'name' of a network and is used to separate different networks.
- Without knowing the ESSID (and assuming no hacking) it should not be possible to participate in the WLAN.
- The architecture of the distribution system is not specified further in IEEE 802.11.
- It could consist of bridged IEEE LANs, wireless links, or any other networks.
- Stations can select an AP and associate with it.
- The APs support roaming (i.e., changing access points), the distribution system handles data transfer between the different APs.
- APs provide synchronization within a BSS, support power management, and can control medium access to support time-bounded service.

SYSTEM ARCHITECTURE

Ad-hoc network architecture

- In addition to infrastructure-based networks, IEEE 802.11 allows the building of ad-hoc networks between stations.
- Every ad-hoc network thus forming one or more independent BSSs (IBSS) as shown in Figure.
- In this case, an IBSS comprises a group of stations using the same radio frequency.
- Stations STA_1, STA_2, and STA_3 are in $IBSS_1$, STA_4 and STA_5 in $IBSS_2$.
- This means for example that STA_3 can communicate directly with STA_2 but not with STA_5.



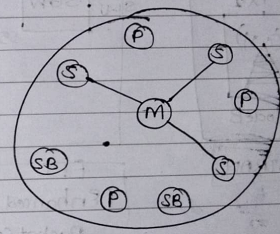
- In this case, an IBSS comprises a group of stations using the same radio frequency.
- Stations STA1, STA2 and STA3 are in IBSS1 and STA4, STA5 are in IBSS2.
- This means for example that STA3 can communicate directly with STA2 but not with STA5.

Q] Describe Bluetooth Architecture.

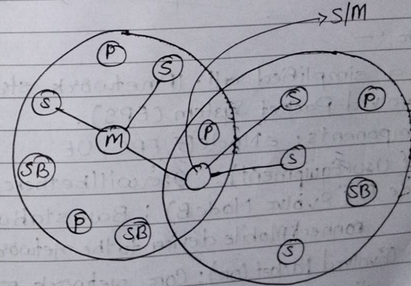
→ There are 2 types of networks made from Bluetooth:
1. Piconet and
2. Scatternet.

1. Piconet :—
- Piconet is a collection of Bluetooth devices which are synchronized to the same hopping sequence.
- Each piconet has only one device called Master (M). All other devices called Slaves (S) are connected to the Master.
- A Master (M) can connect upto 7 active Slaves and upto 255 parked slaves per piconet.
- The Master determines the hopping sequence in the piconet and all slaves have to synchronize to this pattern. If a device wants to participate it has to synchronize to this.
- There are 2 more types of devices
  i) Parked devices (P): cannot actively participate in the piconet but are known & can be reactivated within milliseconds
  ii) Standby (SB) devices: donot participate in the piconet.
- All active devices uses same hopping seq. & hops together.
- All active devices are assigned = 3 bit Active Member Address (AMA)
- All parked devices are assigned = 8 bit Parked Member Address (PMA)
  3 bit AMA
  8 bit PMA

2. Scatternet :—
- Scatternet is a group of piconets. More than one piconet can be connected to form a scatternet through the sharing of common master or slave devices.
- Devices can be slave in one piconet and master in another.
- communication betn piconets can take place by jumping devices back & forth the piconets.



a) Piconet.

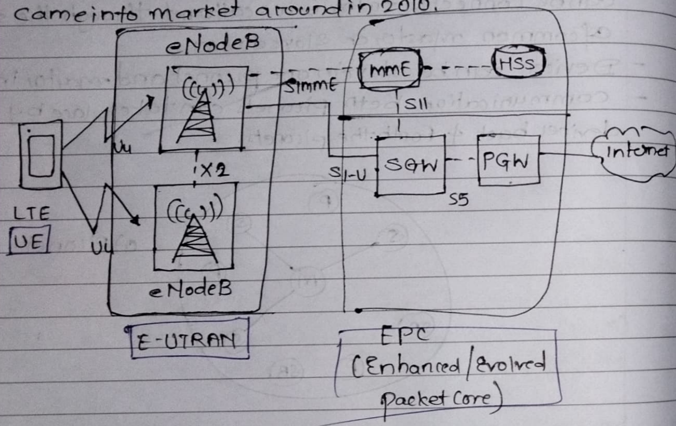


b) Scatternet.

Q] Write a shortnote on: LTE.

→

- Long Term Evolution (LTE) is a 4G wireless broad band technology developed to improve mobile data speed capacity and reliability compared to 3G network.
- It was standardized by 3GPP (third generation Partnership Project)
- Came into market around in 2010.



I] Architecture :-

- LTE uses a simplified all-IP network structure called Evolved Packet System (EPS).
- Main components: eNodeB, EPC, UE.
  - i) UE (User Equipment) :- User will be there device mobile
  - ii) eNode B (Evolve Node B) : Base station that connect Mobile devices to the network.
  - iii) EPC (Evolved Packet Core): Core network responsible for data routing, authentication, mobility mgmt
- This flatter Architecture reduce delay and improves data transmission efficiency.

II] Technical feature :-

- Bandwidth flexibility - supports 10MHz to 20MHz channel bandwidth.
- Peak speeds -
  - ★ Downloads upto 100 Mbps.
  - ★ Upload upto 50mbps.
- Low latency - Around 10 ms, much lower than 3G
- Duplex Modes -
  - ★ FDD (Frequency Division Duplex)
  - ★ TDD (Time Division Duplex)
- Advanced Antenna System - MIMO technology increase data rates and signal quality.

III] LTE - Advanced :-

- Speed upto 1Gbps.
- Carrier aggregation.
- Enhanced MIMO support.
- Better coverage and capacity.

LTE - Adv meets the reqrment of true 4G standard defined by the ITU.

IV] Advantage :-

1) Higher dataspeed.
2) Better voice quality with VoLTE.
3) Improved coverage
4) Efficient spectrum usage.
5) Lower operating cost for service providers.

Q] Discuss in detail about Wifi security protocol.

- Wifi-security protocols are standards used to protect wireless networks from unauthorized access, data theft and attacks.
- They ensure confidentiality, Integrity and Authentication of data transmitted over wireless networks.

I] WEP (Wired Equivalent Privacy):

- Introduced : Early Wifi standard
- Security level : Very weak (obsolete)

Features :

- Uses RC4 encryption algorithm
- Same key used repeatedly
- Simple authentication Mechanism

Limitations :

- Easily cracked due to weak encryption.
- Not recommended for modern use.

II] WPA (Wifi Protected Access):

- Introduced : As an improvement over WEP.
- Security level : Moderate (now outdated).

Features :

- Changes keys frequently.
- Supports Message Integrity check.

Limitations :

- Still vulnerable to certain attacks.
- Not secure enough for modern attacks.

III] WPA2 :

- Introduced : 2004.
- Security level : Strong (widely used)

Features :

- Uses AES (Adv. Encryptn Standard).
- Provides strong encryption and data integrity.

Limitations :

- Vulnerable to KRACK attack if not updated.
- Weak passwords can still be exploited.

IV] WPA3 :

- Introduced : Latest standard.
- Security level : Very strong

Features :

- Provides forward secrecy (past data remains safe even if password is leaked).
- Strong protection against Brute-force attacks.

Limitations :

- Requires newer h/w support.
- Not yet universally adopted.

Protocol	Security Level	Encryption	Status
WEP	Very weak	RC4	Obsolete
WPA	Moderate	TKIP	Outdated
WPA2	Strong	AES	Widely used
WPA3	Very strong	AES + SAE	Latest